

1. An organization called Chifundo International ensures that its IT requirements and implementation are managed. It also ensures that processes are planned, performed, measured and controlled.
 - a. Name the maturity level according to CMMI (Capability Maturity Model Integration). [1]
 - b. Describe any one higher maturity level the organization may move to if process improvement is introduced. [3]
2. Enterprise Architecture principles provide guidance that relates to architecture efforts.
 - a. Describe any two items that influence development of architectural principles. [4]
 - b. What two characteristics should a good set of principles have? [4]
3. Describe any two viewpoints as per the ISO Reference model for Open Distributed Processing (RM-ODP). [4]
4. Glory Suppliers is a company that supplies commodities to its customers. Two of its departments: Sales and Distribution work hand in hand. The Sales department receives an order which triggers a process. The Sales department authorizes the payment by visa card. This is followed by visa card authorization in an external financial institution. The financial institution sends feedback to the Sales department. The Sales department then processes the order. Next, the ordered goods are packed in the Distribution department. Finally, the Distribution department ships the goods.
 - a. Represent this information in a business process diagram showing the message flow between the two departments and the external financial institution. [10]
 - b. Mention one abstract process from the information above. [2]
 - c. Provide one advantage of using BPMN notation in (5a). [2]
5. Describe two types of compliance with UML. [4]
6. Provide two criteria for selecting EA tools. [3]
7. State two advantages of selecting a single EA tool and one disadvantage.
8. Chikanda enterprise has been using a traditional approach to software design since its establishment in 1988. They are planning to transition to an approach to software design in which applications are assembled from reusable components (called services). [1]
 - a. What is the name of this approach? [6]
 - b. Describe three benefits of using the approach described above. [6]
 - c. Provide two challenges they may encounter during deployment using this approach. [4]

1. Describe any three management areas and how they are relevant to Enterprise Architecture. [6]
2. With the aid of a well-labelled diagram, explain how organizations can use enterprise architecture as a management tool to achieve their mission or goal. [10]
3. Compare and contrast the following:
 - a. Any two methods of IDEF description language. [6]
 - b. BPML and UML [4]
4. Why are languages for enterprise modelling significant in EA. Give two points [4]
5. Tachira clinic is a reliable health facility in Matawale. They have consulted an information systems consultant who plans to use Business Process Modelling Notation to represent the process of occurrence of illness in a patient to receiving treatment after consulting a medical doctor at the clinic. The IS consultant comes up with two points of view: one for the doctor and one for the patient.
 - a. Provide the business process diagram showing the two viewpoints and how they interact. [10]
 - b. Write a short paragraph to the clinic discussing the significance of their choice of business process modelling over other alternatives. Give two points. [6]
6. Service Oriented Architecture (SOA) is one of the approaches introduced to solve problems in the IT industry.
 - a. Describe any two problems that SOA addresses. [4]
 - b. Describe any two architectural principles of SOA. [4]
7. Mapale wood furniture is looking into expanding their services in Malawi. One way of achieving this is through venturing into selling fabrics commonly used for making furniture. You are the enterprise architect of this firm.

Using the CSVLOD Model,

- a. Come up with a solution overview outlining the solution and the goal, scope and stakeholders, requirements, expected business benefits and key risks involved. [15]
- b. Describe the relationships between the different types of artifacts in the model. [10]

Mwayi enterprises would like to incorporate a collection of special documents (artifacts) to describing various aspects of their organization from an integrated business and IT perspective. This requires involving **various stakeholders**; **choosing a framework** to use after considering its advantages and disadvantages, ensuring there is **alignment of relevant components** and **selecting areas in the firm where to develop the architecture**. Steps of the architecture **description life cycle** are to be followed in the process. **Four perspectives** are available, based on Balanced Scorecard, on how the business should be viewed. A number of **factors** have motivated the organization to develop these artifacts. If the enterprise does not develop these special documents, a couple of risks are likely to follow.

Write an essay discussing the case study above. In your discussion, articulate on all the phrases in bold. Your essay should not be more than two pages. [25]

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